

Cytomove: a browser-local and reviewable workflow for scratch wound healing assay quantification

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Abstract

The in vitro scratch wound healing assay is one of the most widely used methods for studying collective cell migration, but converting assay images into reproducible measurements remains a practical bottleneck of manual tracing, local software installation, parameter bookkeeping, and limited visibility into how the wound region was segmented. We present Cytomove, a browser-local software tool for reviewable scratch wound healing assay quantification. Cytomove imports local microscopy images, segments the wound region with an explainable variance-and-threshold image-processing pipeline implemented entirely in client-side JavaScript, displays the segmentation as an overlay for user review, supports single-image and grouped time-course analysis, and exports tables, plots, overlay images, and full analysis metadata. Because all processing runs in the browser or in a desktop package built on the same code, microscopy images never leave the user's machine. In a preliminary comparison with the established ImageJ/Fiji Wound Healing Size Tool (WHST) across five image sets and 31 paired measurements, Cytomove reproduced wound-area behaviour closely in clean (mean absolute percentage error 4.1%, Pearson $r = 0.9975$) and phase-contrast time-course images (median area error 6.6%, $r = 0.9984$), while surfacing near-closure and real-world difficulties through overlays and quality-control labels. Cytomove lowers installation friction, keeps data local, and links every exported number to the segmentation image and parameters that produced it.

Keywords

scratch assay; wound healing assay; cell migration; image analysis; browser-based analysis; segmentation; reproducibility

Introduction

The in vitro scratch wound healing assay is a convenient and inexpensive method for studying collective cell migration. A scratch or gap is introduced into a confluent monolayer of adherent cells, the gap is imaged over time, and the rate at which cells move into the cell-free region is used as a proxy for migratory behaviour. The assay remains popular because it is technically accessible, requires no specialised reagents, is compatible with a wide range of adherent cell types, and produces an intuitive readout of coordinated cell movement (Liang et al., 2007;

Jonkman et al., 2014; Grada et al., 2017), a process central to morphogenesis, tissue repair, and cancer invasion (**Friedl and Gilmour, 2009**). It is used routinely in cancer biology, wound healing, vascular biology, and drug-screening contexts.

The simplicity of the wet-lab procedure contrasts sharply with the difficulty of the image analysis. Manual measurement of wound area or width is slow and subjective, and it scales poorly when an experiment contains many fields, time points, replicates, or treatment arms. General-purpose platforms such as ImageJ and Fiji (**Schneider et al., 2012; Schindelin et al., 2012**), and dedicated bioimage-analysis environments such as CellProfiler (**Carpenter et al., 2006; Stirling et al., 2021**) and ilastik (**Berg et al., 2019**), are powerful and widely adopted, but scratch-assay users must still install plugins or macros, choose parameters, inspect the resulting masks, and keep a record of the settings that produced each measurement. In practice, these settings are frequently lost, which undermines reproducibility even when the underlying numbers are sound.

Several dedicated tools have shaped current expectations for scratch-assay software, and each contributes a design lesson. TScratch introduced an automated wound-detection workflow paired with visual inspection and optional manual correction, making the point that segmentation should remain editable and inspectable rather than hidden (**Gebäck et al., 2009**). The ImageJ/Fiji Wound Healing Size Tool (WHST) provided a local-variance and thresholding plugin that reports wound area, area fraction, average width, and width variability, establishing a measurement vocabulary that has become a de facto standard (**Suarez-Arnedo et al., 2020**). PyScratch demonstrated a Python graphical interface for batch and time-course analysis for non-programmer users (**Garcia-Fossa et al., 2020**). More recently, CSMA emphasised the detection of cells located inside the wound region and the production of visual output images (**Pham et al., 2025**). Other automated approaches have likewise combined texture-based segmentation with kinematic readouts, including high-throughput bright-field assays (**Zordan et al., 2011**) and objective continuous-kinematics curve fitting (**Topman et al., 2012**).

These tools also leave practical, workflow-oriented gaps: they require local software environments, depend on manual parameter tracking, and are less naturally connected to browser-based sharing, review, and export. The remaining opportunity is therefore not a new biological metric but lower installation friction, clearer provenance, easier group review, explicit linkage between the exported number and the segmentation that produced it, and the ability to keep unpublished microscopy on the researcher's own machine. Cytomove was developed around this opportunity. It is a browser-local software tool that treats the segmentation mask as a review object rather than a hidden intermediate, and records the parameters, crop, orientation, and correction state behind every exported measurement.

The field is also moving beyond simple endpoint closure measurements. Recent work has addressed the reproducibility of scratch creation itself through low-cost robotic scratching platforms (**Lin et al., 2024**), and spatially resolved analysis frameworks now combine wound-edge detection with local cell-trajectory or sector-based measurements to connect global closure

with regional migration behaviour (Vašínková et al., 2025). These developments reinforce the same practical requirement: scratch-assay software should not only output a number, but should also make the image evidence, segmentation assumptions, and analysis context visible.

Materials and Methods

Software implementation and workflow

Cytomove is a browser-based scratch wound healing analysis tool. The user opens the web application, loads one or more local microscopy images, selects or adjusts an image-type preset, reviews the segmentation overlay, optionally applies local corrections, and exports the results. All image processing is performed locally in the browser using the HTML Canvas API and typed-array operations; no external image-processing backend is required, and assay images are not transmitted to a server. A desktop package built on the same analysis code is also available for heavier local workflows and more predictable performance on large images.

Software architecture

The Cytomove segmentation engine is an explainable, dependency-free image-processing pipeline implemented independently in JavaScript using typed arrays, without ImageJ/Fiji, WHST, OpenCV, or other external image-processing backends. WHST is used in this manuscript only as a quantitative comparator because it reports a closely related measurement vocabulary; the WHST plugin or code is not invoked by Cytomove. The pipeline proceeds through the following stages:

1. **Grayscale conversion.** The RGB image is converted to luminance using the ITU-R BT.709 / sRGB weighting ($0.2126R + 0.7152G + 0.0722B$).
2. **Field-of-view masking.** An optional field-of-view cutoff masks black corners and circular microscope-field borders so that only the imaged field contributes to the analysis.
3. **Contrast enhancement.** Pixel intensities are clipped to the 1st and 99th percentiles and normalised, stabilising the subsequent variance and threshold steps against outliers and uneven exposure.
4. **Local variance map.** A local variance is computed for every pixel within a user-selected radius using an integral image (summed-area table) (Crow, 1984; Viola and Jones, 2001), so that per-pixel variance is obtained in constant time regardless of radius. Confluent cell regions have high local texture variance, whereas the cell-free wound corridor is comparatively smooth.
5. **Thresholding.** The variance map is thresholded to separate the low-variance wound from the high-variance cell sheet. Otsu's method (Otsu, 1979) is used by default; when the Otsu threshold collapses toward zero on low-contrast images, a percentile-based fallback is applied and recorded in the export.
6. **Island filtering.** Connected components below a user-set minimum size are removed before hole filling, suppressing isolated speckles and debris.

7. **Hole filling.** Holes inside the wound region are filled using a border flood-fill and inversion procedure, a standard binary morphology operation (Soille, 2003), so that small cell clusters inside the gap do not punch holes in the wound mask.
8. **Continuity and component selection.** Connected components are scored by central proximity, axis span, and area, and the wound-like continuous component is retained; additional steps bridge nearby fragments, extend a stable corridor to the image frame where the wound continues out of frame, and, for phase-contrast images, close narrow internal slit artefacts.
9. **Width estimation.** For each image row, the horizontal span of the wound is measured after the selected scratch orientation and any fine rotation have been applied. Cytomove reports the mean, median, standard deviation, and coefficient of variation of the row-wise widths, together with the number and fraction of valid rows.

The tool ships with image-type presets (brightfield normal cells, brightfield small cells, phase contrast, and a speckled phase-contrast variant) that set sensible defaults for variance radius, threshold scale, minimum component size, island handling, field-of-view cutoff, and microscope mode. When a group is loaded, Cytomove samples the images and selects an initial preset automatically; manual preset or microscope-mode changes are treated as user overrides.

Software functionalities

The tool supports two analysis modes. In single-image mode, one image is displayed in a review canvas with its segmentation overlay and current measurements, and the user can adjust parameters and corrections. In group mode, multiple images (for example a time course or treatment set) are loaded together, reviewed frame by frame through thumbnail cards, and exported as group-level tables, plots, and overlay files; each image retains its own segmentation settings.

The segmentation output is displayed as a contour overlay on top of the source image rather than as a separate binary mask, letting the user inspect — before exporting any number — whether the wound corridor was captured correctly, whether cells inside the wound were treated appropriately, whether circular field borders were excluded, and whether near-closure regions remain biologically meaningful. When automatic segmentation is insufficient, Cytomove provides local manual-correction tools that operate on a dragged rectangular region of interest: a scan-based Add mode that runs a local fine threshold and writes back the largest connected gap component; a direct Fill mode; an Erase mode; and a Clean specks mode. Corrections persist across navigation, and the correction status of each image is stored in the export metadata so that manually adjusted measurements are never silently mixed with fully automatic ones.

For every analysed image, Cytomove exports wound area in pixels, wound area fraction, mean and median wound width, width standard deviation and coefficient of variation, valid-row count and fraction, quality-control labels and warnings, the recommended primary metric, crop and rotation state, the full set of segmentation parameters, manual-correction status, image

dimensions, the algorithm version string, and source metadata where available. These fields make each measurement auditable after the fact. Exports are provided as CSV and Excel tables, PNG overlay images, grouped overlay ZIP archives, and time-course plot images. Figure 1 shows the application workspace, and Figure 2 shows representative overlays across three imaging contexts.

Validation datasets and comparator analysis

Cytomove was evaluated against WHST across five image sets comprising 31 paired measurements, including two 11-frame time-course sequences (Table 2). WHST was selected as the primary quantitative benchmark because its ImageJ/Fiji output vocabulary overlaps with Cytomove's exported endpoints. For each dataset, the signed and absolute percentage difference between Cytomove and WHST was computed for area and width, and summarised as the mean absolute percentage error (MAPE), the median absolute percentage error, the maximum absolute percentage error, the mean signed error, and the Pearson trend correlation (Table 3). MAPE and median error are reported together because, in time courses that approach closure, the residual wound becomes very small and a tiny absolute difference can produce a large percentage error in a single frame.

Results

The validation was designed to ask a practical question: whether a browser-local, reviewable implementation can reproduce the wound-area behaviour of an established ImageJ/Fiji comparator while making difficult frames visible to the user. The five sets therefore combine a clean public comparator sequence, a public phase-contrast time course that approaches closure, and three author-acquired phone-capture sets used as external support or stress material rather than as definitive accuracy claims. The numerical summary is reported in Table 3, and each main result is paired with visual overlays or audit panels in Figures 2-6.

In the clean comparator sequence, the `csma_sample_11` subset, an 11-frame sequence drawn from the CSMA WoundHealing sample data (**Pham et al., 2025**), provided the clearest support: Cytomove wound-area measurements agreed closely with WHST (area MAPE 4.1%, median 3.9%, maximum 7.9%, mean signed error -2.7% , Pearson $r = 0.9975$); width agreement was acceptable (MAPE 9.7%, $r = 0.9969$). A three-method comparison was possible because native CSMA, WHST, and Cytomove area outputs were available for the matched frames (Figure 5). All three followed the same decreasing wound-area trajectory (CSMA–WHST $r = 0.9970$; CSMA–Cytomove $r = 0.9951$), while native CSMA retained larger residual areas in later frames, presented as method-context evidence rather than a failure case.

In the phase-contrast WHAD-MCF7 time course, the `whad_mcf7_11` set, an 11-frame phase-contrast MCF-7 sequence from the publicly deposited WHAD/CAMAD dataset under CC BY 4.0 (**Iheme et al., 2024**), tested Cytomove on a wound that progresses to near-complete closure (Figure 6). Cytomove followed the WHST area and width trends closely ($r = 0.9984$ and $r = 0.9986$) with a modest median absolute area error of 6.6% and a small positive mean signed error

(+2.2%). The dataset-level area MAPE of 15.0% was concentrated in the latest near-closed frames: in the final frame the residual wound measured roughly 1,400 pixels by WHST versus roughly 2,550 pixels by Cytomove, an absolute difference of just over a thousand pixels that nonetheless registers as an ~82% relative error. Cytomove flags such frames with quality-control labels (`review_needed`, `area_ok_width_review`) and exposes the overlay, valid-row information, and width variability.

In the smartphone-through-eyepiece stress sets, three smartphone-through-eyepiece brightfield sets acquired by the author (HK, M8F, MK; three frames each) reproduce conditions common in real laboratories but absent from curated datasets: circular fields, uneven illumination, focus variation, and fragmented late wounds. The `cell_phone_HK` set behaved as a supportive external example (area MAPE 6.7%, $r = 0.9903$); `cell_phone_M8F` showed a late-frame underestimation pattern (area MAPE 10.6%, mean signed error -10.6%); and `cell_phone_MK` behaved as a genuine limitation case in which Cytomove over-measured fragmented near-closed regions (area MAPE 38.8%, width MAPE 31.0%), with every difficult frame flagged for review. These sets are reported as quality-control and stress examples rather than solved-accuracy evidence.

Taken together, a pooled Bland–Altman comparison (**Bland and Altman, 1986**) on the absolute area scale (Figure 3B) showed a median Cytomove-minus-WHST difference of essentially zero (-0.002 Mpx), with 2.5–97.5th percentile limits of agreement of -0.134 to $+0.169$ Mpx; an absolute, non-parametric summary is used because a percentage difference is unstable for near-closure frames and because the pooled points include time-course frames that are not mutually independent. Across all sets, area was the more stable and comparable metric; width tracked area but was more sensitive in fragmented and near-closed frames (Figure 4; Supplementary Figure S1), supporting the area-first, width-as-quality-control interpretation adopted in this work.

The application-level figure establishes the workflow context for these comparisons. Figure 1 shows that Cytomove presents parameter controls, wound-overlay review, quality-control labels, and frame-level group navigation in the same workspace. This matters for interpreting the quantitative results because the exported measurements are not isolated table entries: they are linked to the mask, crop, rotation, preset, manual-correction status, and warnings that produced them. Figure 2 then illustrates the range of image appearances used in the evaluation. In the clean CSMA image the wound corridor is visually continuous and strongly separated from the cell sheet; in the WHAD-MCF7 phase-contrast example the contrast is softer and the wound approaches late closure; and in the phone-capture example the circular field and acquisition artefacts create a more difficult segmentation context. These examples are included to make clear that the numerical validation was not treated as a black-box benchmark detached from the underlying images.

The paired area plot and Bland–Altman panel in Figure 3 provide complementary views of agreement. The identity-line view shows that most paired wound-area measurements fall close to the WHST reference across the clean and phase-contrast sequences, while the phone-capture limitation set contributes the more visibly dispersed points. The Bland–Altman view is

intentionally reported on the absolute area scale rather than as a percentage difference because near-closure frames are numerically unstable: once the remaining wound is only a few thousand pixels, a small absolute offset can dominate a relative-error statistic. The median Cytomove-minus-WHST difference was essentially zero, and the 2.5-97.5th percentile range remained narrow relative to the full range of wound areas represented in the time courses. This supports the interpretation that Cytomove is able to follow the main wound-area trajectory in suitable images, while also showing why individual late frames require visual review rather than automatic acceptance.

Figure 4 connects the numeric summaries with visual evidence. The representative WHST and Cytomove outputs show that agreement is strongest when the wound corridor is continuous, centrally located, and visually separable from the cell sheet. Conversely, the more difficult cases show why a reviewable overlay is not a cosmetic addition but part of the measurement workflow. In fragmented late wounds, a single scalar endpoint can suggest high correlation across a time course even though one or more frames need manual inspection. Cytomove's quality-control labels and valid-row metrics therefore serve as a companion to the numerical output rather than as a post hoc warning.

The three-method CSMA comparison in Figure 5 adds useful context because it places Cytomove and WHST beside native CSMA output for the same matched sequence. All three methods captured the monotonic decrease in wound area, and the pairwise correlations among CSMA, WHST, and Cytomove were high. The main difference was not the direction of the trajectory but the residual area retained by native CSMA in later frames. This pattern is reported as method-context evidence: different tools can preserve the same biological trend while making different choices at the wound boundary, especially when residual gaps become small. Cytomove's contribution here is that the choice is visible in the overlay and traceable in the exported metadata.

Figure 6 focuses on the WHAD-MCF7 phase-contrast sequence, where the same issue is more pronounced because the wound approaches near-complete closure. The area and width trajectories remained strongly correlated with WHST, but the mean area error was pulled upward by the latest frame. The median area error gives a more representative summary of the sequence, while the maximum error identifies the review-sensitive endpoint. This is why Table 3 reports mean, median, maximum, and correlation together. No single metric is sufficient: correlation captures trend agreement, MAPE captures average relative discrepancy, median error reduces the influence of the near-closure tail, and the maximum error identifies the frame that a user should inspect.

Discussion

The main finding is that Cytomove can reproduce WHST-like wound-area behaviour in image sets that are visually suitable for variance-and-threshold segmentation, while preserving enough visual and metadata context to identify frames that should not be interpreted automatically. The

clean CSMA sequence provides the strongest agreement evidence, with low mean and median area error and near-unity area and width correlations. The WHAD-MCF7 sequence supports the same direction of evidence in phase-contrast microscopy, but it also demonstrates why near-closure frames need to be reported carefully: small residual wounds turn modest absolute differences into large percentage errors. The phone-capture sets broaden the evaluation by showing how circular fields, uneven illumination, focus variation, and fragmented late wounds appear in the Cytomove quality-control workflow.

Cytomove is a workflow and reproducibility contribution rather than a new biological metric. The established scratch-assay tool literature already defines the major measurement concepts (wound area, normalised area, width, closure behaviour, time-course plots, output images, and spreadsheet export); Cytomove brings these into a browser-local, reviewable workflow that reduces installation friction while preserving inspectability and recording the provenance of each measurement. To reproduce a measurement, a researcher needs to know which image was analysed, which crop or field was used, what threshold and variance settings were applied, whether rotation or manual correction occurred, and which software version produced the output — and Cytomove records all of these in its exported metadata, directly addressing the common loss of analysis settings in macro- and plugin-based workflows.

The local-first design is equally consequential. Browser-local analysis lowers the barrier to trying the workflow and supports privacy-sensitive or unpublished microscopy that researchers may be unwilling to upload to a remote service, while the desktop package provides more predictable performance for large experiments. By keeping the segmentation overlay linked to every exported number, Cytomove makes measurement uncertainty visible: strong trend concordance with WHST and review-sensitive individual frames can coexist, and the tool surfaces both rather than collapsing a time course into a single pass/fail number. Cytomove is therefore most useful to cell biology, cancer biology, and drug-screening laboratories that run scratch assays routinely and need an accessible, auditable, publication-oriented analysis path that complements rather than replaces ImageJ/Fiji.

This emphasis on provenance is not merely a software-engineering preference. Scratch-assay measurements are often used downstream as evidence for treatment effects, gene perturbation, drug response, or phenotypic differences in migration. If the exported number is separated from the mask and settings that generated it, the reader cannot tell whether a reported difference reflects cell movement, a threshold choice, a crop boundary, an unrecognised field-of-view artefact, or a manual correction. Cytomove therefore treats the segmentation as an object of review. This aligns with the older lesson from TScratch that automated detection should remain visually inspectable, with WHST's measurement vocabulary for area and width, with PyScratch's emphasis on usability for non-programmers, and with CSMA's reminder that cells inside the wound region can materially alter how a gap should be interpreted.

Cytomove does not attempt to replace mature image-analysis ecosystems such as ImageJ/Fiji, CellProfiler, ilastik, or recent deep-learning segmentation tools. Instead, it targets a narrower

workflow gap: researchers who need an accessible scratch-assay analysis path that can be opened in a browser, keeps images local, shows the segmentation overlay before export, and records the settings behind the reported numbers. This positioning is complementary to TScratch, WHST, PyScratch, and CSMA. The present implementation adopts their shared lesson that wound masks must remain inspectable, but places that lesson into a browser-local and export-rich workflow.

The literature around scratch-assay analysis also shows that the experimental problem itself is becoming more spatially explicit. Open-source robotic scratching platforms address one source of variability by making the physical wound more reproducible, while recent level-set and image-sector approaches analyse migration as a spatially resolved process rather than a single global closure fraction. Cytomove is not yet a kinematic or cell-tracking platform, but this direction is important for its interpretation. A practical wound-healing tool should support simple area and width outputs for routine users, while leaving a path toward richer spatial readouts when the image quality and experimental design justify them. The current export schema already records enough per-image context to support that future direction, including the selected preset, crop, rotation, valid-row fraction, quality-control labels, and manual-correction status.

Recent bioimage-analysis literature makes this positioning more, not less, important. Generalist and foundation-model approaches such as Cellpose3, Segment Anything for Microscopy, and MedSAM show how rapidly segmentation methods are advancing across microscopy and medical-imaging domains (**Stringer and Pachitariu, 2025; Archit et al., 2025; Ma et al., 2024**). However, those methods also highlight the need for task-specific validation, appropriate image-domain adaptation, and careful interpretation of out-of-distribution performance. For routine scratch assays, an explainable local pipeline with visible overlays can therefore remain useful even when more powerful segmentation backends become available, because the workflow problem is not only segmentation accuracy but also review, provenance, export, and reproducible reporting.

There is also a pragmatic reason not to make a deep model the default baseline for this first release. Scratch-assay laboratories vary widely in microscope type, illumination, cell density, wound geometry, and image storage practices. A learned model can perform impressively when its training distribution matches the test images, but it can fail silently when the domain shifts. An explainable variance and threshold pipeline is less powerful, but its behaviour is easier to inspect: a smooth low-variance wound region, a textured cell sheet, and a visible contour are concepts that a bench scientist can judge directly on the image. This does not argue against learned segmentation in Cytomove. Rather, it argues that any future learned backend should be optional, reviewable, and evaluated against the same metadata and visual-audit standards used here.

Area is treated as the primary endpoint because it was more stable across the current validation sets and because it maps naturally to the segmented wound mask. Width remains valuable as a secondary quality-control endpoint: it tracks wound closure trends in clean and phase-contrast images, but it is more sensitive to fragmented residual wounds, partial closure, and row-wise

validity. Reporting both area and width therefore helps expose disagreement rather than hiding it inside a single summary statistic.

The treatment of near-closure frames deserves particular caution. In a wound-healing time course, the final biological state may be the most interesting, but it is also where relative error becomes least stable. A difference of roughly a thousand pixels can be small in the context of early open wounds yet large as a fraction of a nearly closed residual gap. This is why the WHAD-MCF7 sequence is useful despite raising the mean MAPE: it demonstrates a realistic failure mode in which trend correlation remains excellent but endpoint relative error becomes large. Reporting both median and maximum error prevents the late frame from being either hidden or overgeneralised. The correct reading is not that Cytomove is uniformly accurate in near-closure images, but that the workflow can identify and expose those frames for review.

The present validation should be interpreted as preliminary. The strongest evidence comes from the clean CSMA sequence and the public WHAD-MCF7 phase-contrast time course, whereas the phone-capture sets are included to expose realistic review and quality-control behaviour rather than to claim solved accuracy under all acquisition conditions. The study does not yet include multi-rater manual consensus masks, Dice or intersection-over-union evaluation, broad cell-line diversity, physical-unit calibration, runtime benchmarking, or a systematic browser-by-browser reproducibility test. The current algorithm also does not explicitly model wound-internal cells, a case highlighted by CSMA and important for future development.

The phone-capture sets should be read with the same restraint. They are valuable because they approximate the imperfect acquisition conditions encountered in many laboratories: eyepiece capture, circular fields, uneven illumination, focus variation, and late-frame fragmentation. They are not a substitute for a larger blinded validation panel. Their role in this preprint is to show that Cytomove can surface stress cases and preserve the evidence needed to decide whether a measurement is usable. A stronger follow-up study should stratify images by modality, cell type, wound geometry, and acquisition quality, and should compare Cytomove not only with WHST but also with expert manual masks and additional dedicated tools.

Planned work includes wound-internal cell handling, validation against multi-rater manual consensus masks with spatial-overlap metrics such as the Dice coefficient and intersection-over-union, spatial and temporal calibration for physical-unit closure rates, runtime and browser reproducibility testing, and reproducible export packages aligned with FAIR data principles. A future optional learned-segmentation backend may be added for difficult images, but the review-first and local-first workflow should remain the organising principle.

Conclusion

Cytomove provides a browser-local and reviewable workflow for scratch wound healing assay quantification. The current evidence supports Cytomove as an accessible, publication-oriented analysis tool that can produce WHST-comparable wound-area outputs in suitable images, while

preserving the overlay and quality-control context needed to interpret difficult near-closure and phone-capture cases.

Ethics statement

This study did not involve human participants, identifiable human data, or animal experiments, and therefore required no ethical approval. The MCF-7 phase-contrast frames were obtained from a publicly available dataset released under the Creative Commons Attribution 4.0 International licence (**Iheme et al., 2024**); the smartphone-through-eyepiece brightfield images were acquired by the author from routine in vitro cell cultures.

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Declaration of competing interest

The author is the developer of Cytomove and intends to offer a commercially licensed version of the software. No other competing interests are declared.

CRedit author statement

Zekeriya Düzgün: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Data and software availability

A live web version is available at <https://cytomove.com>, and the public project repository is <https://github.com/zduzugun/CytoMove>. Public release and licensing of the full core analysis source code are pending institutional IP/TTO confirmation. The preliminary validation summary -- paired WHST-Cytomove measurements, dataset-level diagnostic statistics, the discrepancy-resolution table, and representative overlay images -- is archived on Zenodo under the reserved DOI <https://doi.org/10.5281/zenodo.20486820>. External image provenance: WHAD-MCF7 frames from the WHAD/CAMAD Zenodo dataset under CC BY 4.0 (**Iheme et al., 2024**); CSMA sample images and output context from the public CSMA WoundHealing repository and associated article (**Pham et al., 2025**); smartphone-through-eyepiece images acquired by the author. All Cytomove overlays and analysis outputs were generated by the author.

Software and code metadata

Code metadata for the current preprint version

Nr.	Code metadata description	Value
C1	Current code version	v1.0.0

C2	Permanent link to code/repository used for this code version	https://github.com/zduzgun/CytoMove (public project repository; source-code licensing/public-release confirmation pending institutional IP review)
C3	Permanent link to Reproducible Capsule	— (not applicable)
C4	Legal Code License	Pending institutional IP confirmation; intended open-source core licence under review
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	JavaScript (ES2020+), HTML5 Canvas API, typed arrays; Electron (desktop package)
C7	Compilation requirements, operating environments and dependencies	Runs in any modern Chromium- or Firefox-based browser with no installation; desktop build requires Node.js ≥ 18 and Electron
C8	If available, link to developer documentation/manual	https://github.com/zduzgun/CytoMove (README and /docs)
C9	Support email for questions	zekeriya.duzgun@giresun.edu.tr

A live web version is available at <https://cytomove.com>. Public release and licensing of the core source code are pending institutional IP/TTO confirmation; this preprint does not rely on a completed open-source relicensing step.

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Tables

Table 2. Preliminary validation datasets.

Dataset	n	Imaging / role	Source / provenance	Comparator
csma_sample_1 1	11	Clean comparator time course	CSMA repository sample data / outputs (Pham et al., 2025)	WHST (with CSMA visual context)
whad_mcf7_11	11	Phase-contrast time course	WHAD/ CAMAD Zenodo dataset, CC BY 4.0 (Iheme et al., 2024)	WHST
cell_phone_HK	3	Phone-capture external support	Author-acquired	WHST
cell_phone_M8 F	3	Phone-capture support / stress	Author-acquired	WHST
cell_phone_MK	3	Phone-capture stress /	Author-acquired	WHST

		limitation		
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Table 3. Preliminary Cytomove-versus-WHST agreement summary. Area and width are reported as mean absolute percentage error (MAPE), median absolute percentage error, maximum absolute percentage error, and Pearson trend correlation (r).

Dataset	n	Area MAPE %	Area median %	Area max %	Area r	Width MAPE %	Width r	Interpr etation
csma_s ample_ 11	11	4.1	3.9	7.9	0.9975	9.7	0.9969	Primary clean compar ator; visually concord ant overlay s
whad_ mcf7_1 1	11	15.0	6.6	82.5	0.9984	9.0	0.9986	Phase- contrast ; MAPE driven by late near- closure frames
cell_ph one_H K	3	6.7	3.2	16.3	0.9903	8.5	0.9918	Externa l support set
cell_ph one_M 8F	3	10.6	5.1	24.9	0.9557	8.7	0.9076	Support / stress; late- frame underes timation
cell_ph	3	38.8	43.4	72.3	0.9880	31.0	0.9861	Stress /

one_M K								limitati on case; not accurac y evidenc e
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Table 1 (Code metadata) is provided in the Software and code metadata section above.

Figures

REVIEW MODE

SingleGroup

Loaded group

m8f

BASIC DISPLAY

ContourMask

CONTOUR STYLE

PRESETS

Brightfield normal cellsBrightfield small cellsPhase contrastPhase contrast 2

Apply to group

Auto detect microscope

Parameter controls

BASIC ANALYSIS

Variance radius15

Threshold offset-28

Min component (px)55000

Ignore tiny islandsLow

Microscope modePhase contrastBrightfield

Scratch orientationHorizontal scratch

ADVANCED GEOMETRY

FOV cutoff36

Fine rotation (deg)0

Show angle ruler

Auto crop FOV

Cuts the non-black microscope field before analysis; rectangular crops are allowed.

Apply current crop ratio to group

Uses the same relative rectangle for each image in group review.

Adjust cropApply crop

Reset crop

VIEW

MANUAL CORRECTION

METRICS

EXPORT

m8f_24h_002.png

1800x1800 px horizontal scratch → 90deg variance radius

Overlay review canvas

QC and metric cards

SERIES QC

100/100

crop/FOV looks consistent

LATEST AREA CLOSURE

62,7%

t0 to undefined

LATEST WIDTH CLOSURE

55,1%

t0 to undefined

AREA-WIDTH PEARSON R2

0,997

Pearson correlation squared; metrics track together

RECOMMENDED

area and width

area and width can be interpreted together

m8f_0h_001.png

0h

m8f_24h_002.png

24h

m8f_48h_003.png

48h

Frame-level group review

CM area1,745,326 px

Area %53,87%

CM width980,5 px

Valid rows100%

Image QC97/100

GT closuren/a

CM area1,170,683 px

Area %36,13%

CM width721,1 px

Valid rows100%

Image QC97/100

GT closuren/a

CM area651,852 px

Area %20,13%

CM width440,1 px

Valid rows100%

Image QC100/100

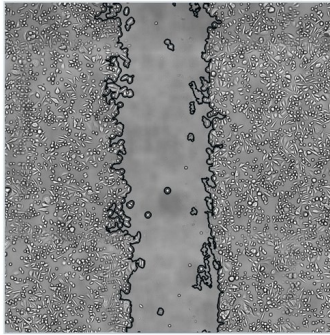
GT closuren/a

Threshold 17 (Otsu 45, offset -28) · radius 15 · group prior r63 · microscope brightfield (3.240.000 px) · wound comps 1/234 · internal islands 702 (1/701) · mask 1.170.683 px · width 721,1 px · 874 contour 12.469 px · 2 QC warnings · GT n/a% area fraction

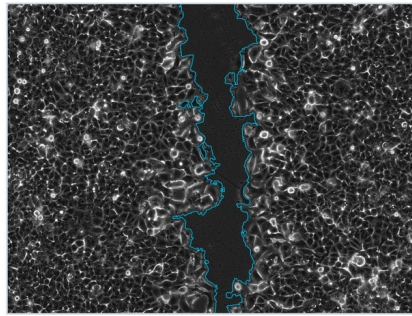
Figure 1. Cytomove application workspace for browser-local scratch assay quantification. The interface keeps adjustable segmentation parameters, overlay-based wound review, quality-control and metric summaries, and frame-level group review in a single workflow before export.

Representative CytoMove segmentation overlays

Clean comparator
CSMA sample



Phase contrast
WHAD-MCF7



Stress case
phone capture



Figure 2. Representative CytoMove segmentation overlays. Overlays for a clean CSMA sample image, a WHAD-MCF7 phase-contrast frame, and an author-acquired phone-capture stress image. The wound contour remains visible so that segmentation quality can be inspected before export.

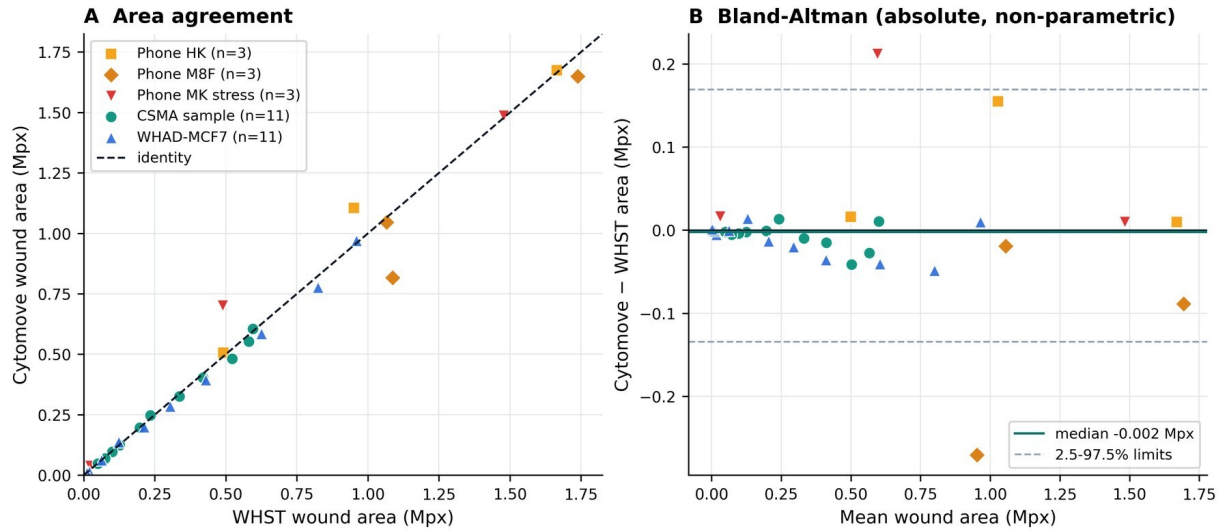


Figure 3. WHST versus Cytomove wound-area comparison. (A) Paired WHST and Cytomove wound-area values against the identity line, with a distinct marker per dataset. (B) Bland–Altman view on the absolute area scale; the solid line marks the median difference (-0.002 Mpx) and the dashed lines the 2.5–97.5th percentile limits of agreement (-0.134 to $+0.169$ Mpx).

Visual agreement between WHST comparator output and Cytomove overlays

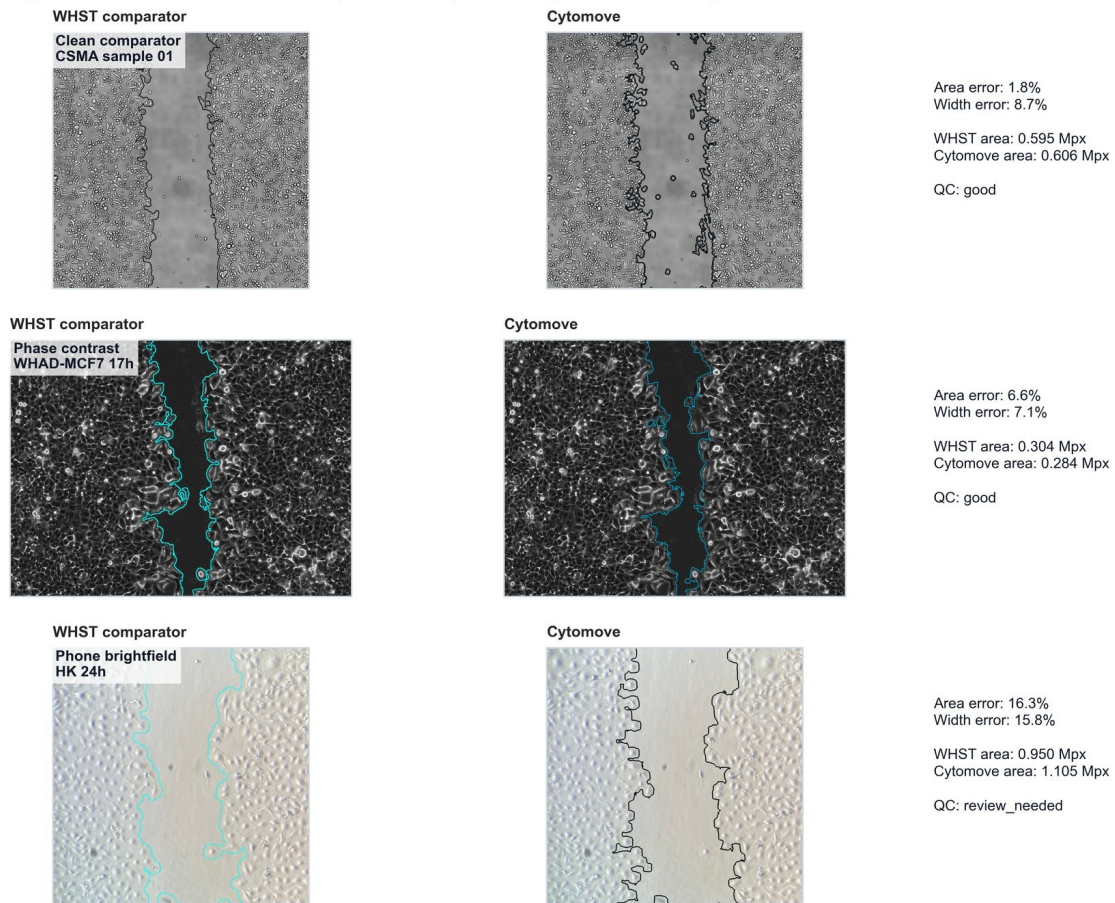
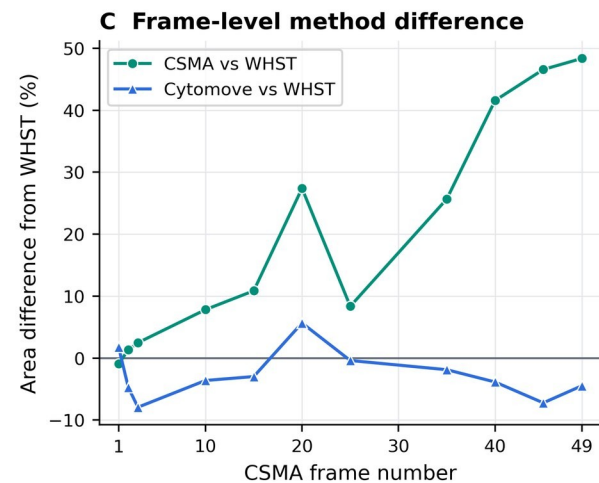
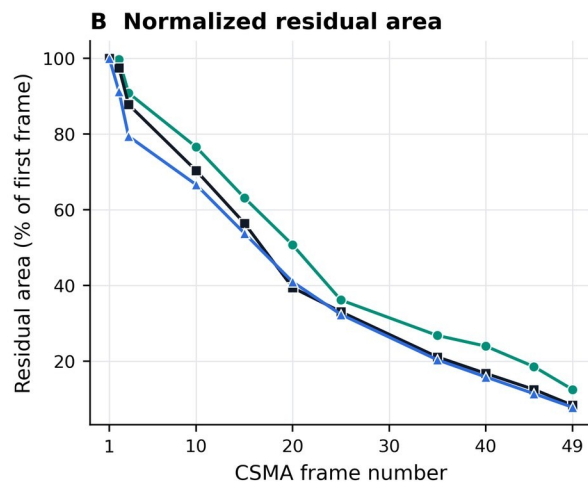
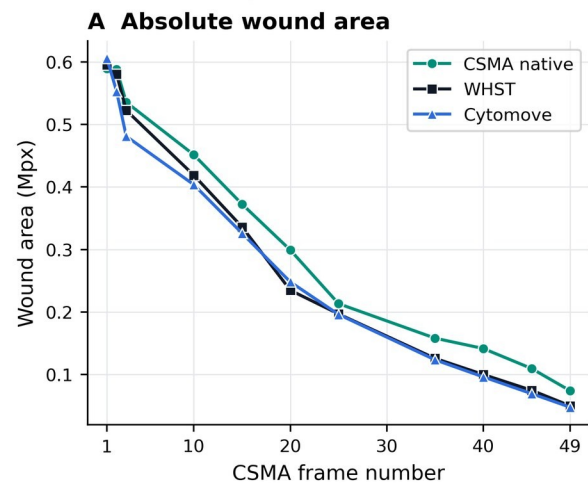


Figure 4. Visual comparison of WHST output and Cytomove overlays. Representative examples connecting numerical comparison with visual evidence across clean CSMA material, WHAD-MCF7 phase-contrast images, and author-acquired phone-capture cases.

CSMA sample 11 three-method area trend



D Frame-level visual audit: raw images and comparator overlays

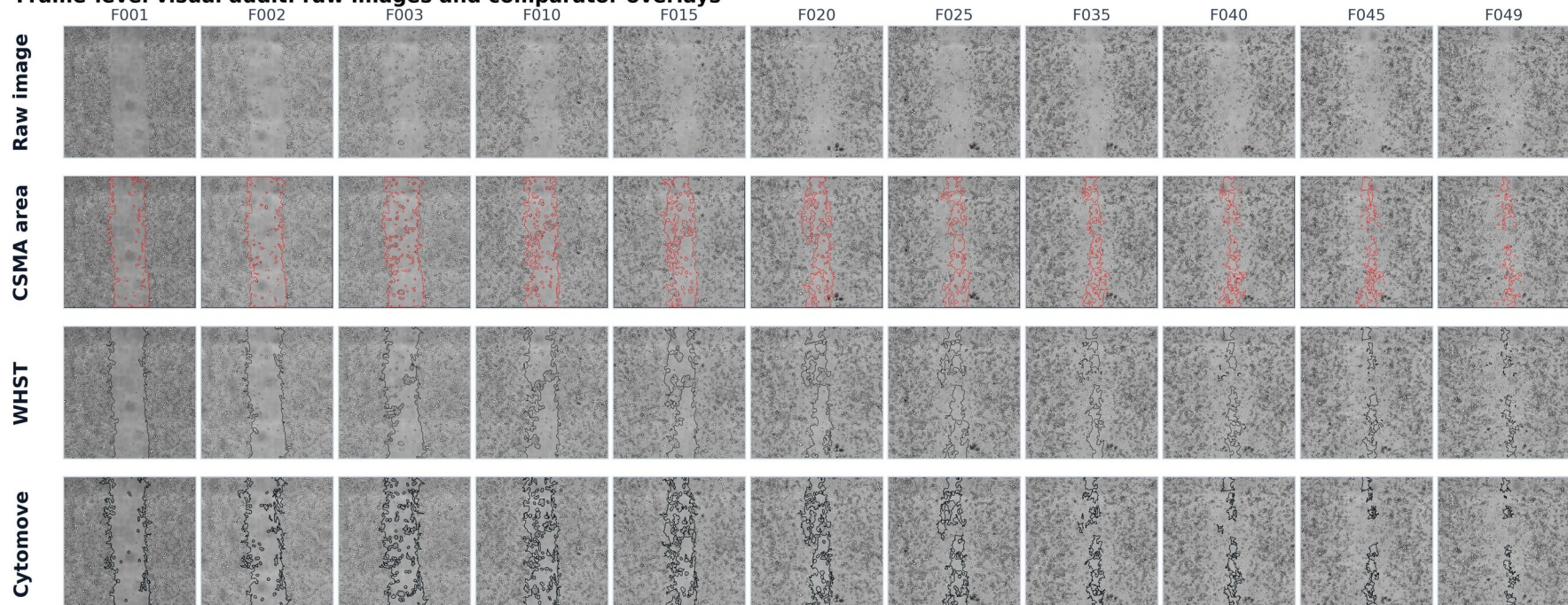


Figure 5. CSMA sample 11 three-method area trend and visual audit. (A–C) Native CSMA, WHST, and Cytomove area across the matched 11-frame sequence; (D) frame-level visual audit showing the raw image, CSMA area output, WHST output, and Cytomove overlay for each frame.

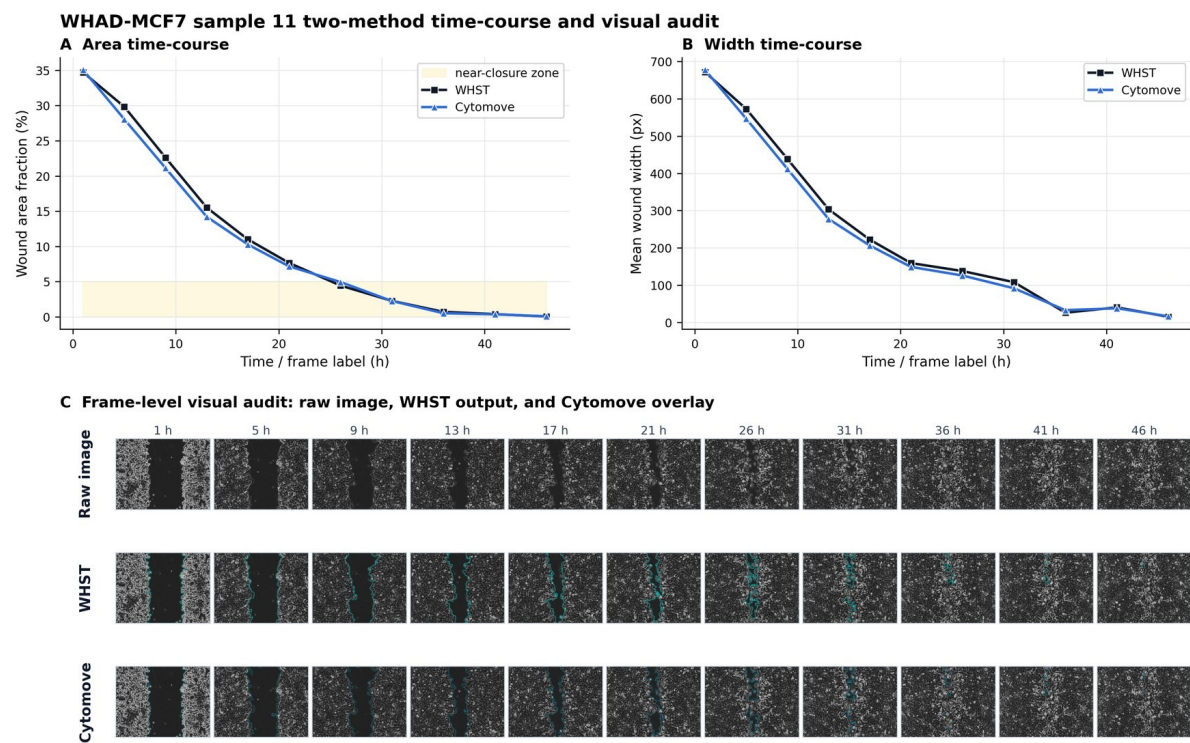
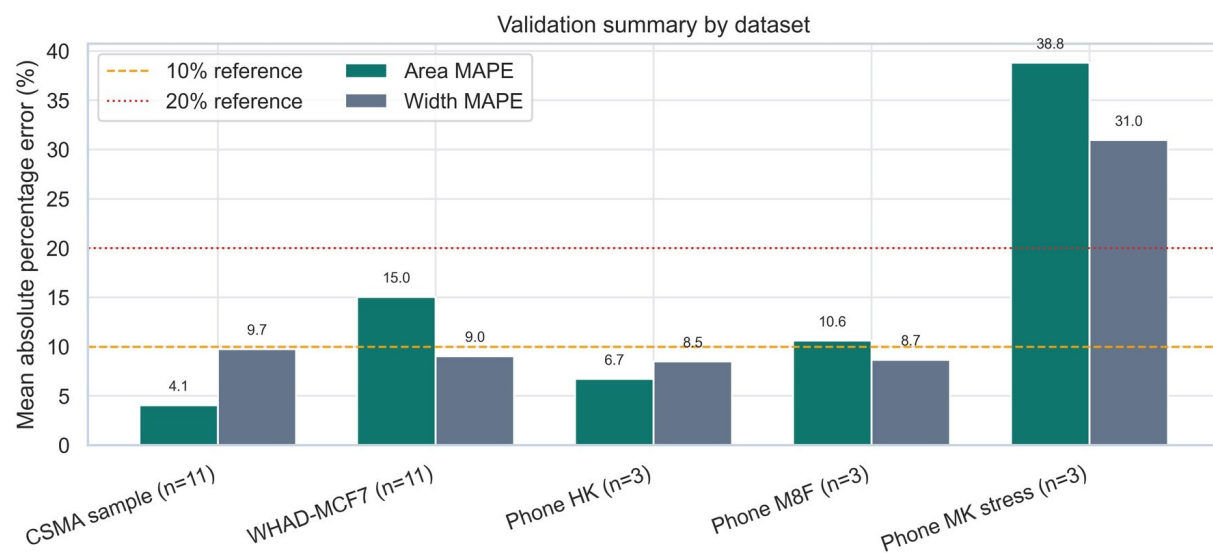


Figure 6. WHAD-MCF7 time-course behaviour and visual audit. (A) wound-area and (B) wound-width values across the 11-frame phase-contrast sequence (Iheme et al., 2024), with (C) a frame-level visual audit; late near-closure frames are flagged as review-sensitive.



Supplementary Figure S1. Dataset-level validation summary. Diagnostic summary of area and width agreement statistics across the five validation sets.